1 Methods

1.1 Model

$$\frac{\mathrm{d}S_i}{\mathrm{d}t} = \sum_{j=1}^{N_P} M_j \left(v_j \cdot \gamma_{ij} - \delta\right) \tag{1}$$

$$v_j = v_{\max,j} \cdot f_{\mathrm{MM},j} \cdot f_{\mathrm{thermo},j} \cdot f_{\mathrm{inhib},j} \tag{2}$$

$$f_{\mathrm{MM},j} = \prod_{i=1}^{N_\mathrm{S}} \frac{S_i}{S_i + K_{m,ij}} \tag{3}$$

$$f_{\mathrm{inhib},j} = \prod_{i=1}^{N_\mathrm{S}} \frac{S_i}{S_i + K_{inhib,ij}} \tag{4}$$

If the i-th pool is irrelevant for the j-th pathway, $K_{m,ij} = 0$. Numerics: Add $\varepsilon = 10^{-12}$ in the denominator, to avoid division by zero.

$$S_i^{\mathrm{dis}} = \tag{5}$$

$$f_{\mathrm{thermo},j} = 1 - e^{\frac{\min(0, \Delta G_r - \Delta G_{r,\min})}{RT}} \tag{6}$$